Monitoring data and models 30
Applying our learning 31
Technical requirements 31
Setting up your workspace 32
Kaggle setup 32
Starting the projects 40
Project: Favorita Store Sales – Time Series Forecasting 40
Project: Streaming Transactions 41
Project: Retrieval-Augmented Generation Chatbot 44
Project: Multilabel Image Classification 45
Summary 46
Questions 46
Answers 46
Further reading 46

3

Building the Bronze Layer 49

Revisiting the Medallion architecture pattern 49
Transforming data to Delta with Auto Loader 50
Schema Evolution 51
Delta Live Tables for the Bronze layer 51
Delta Live Tables benefits and features 52
Bronze data with DLT 53
Maintaining and optimizing Delta Tables 53
Vacuum 54
Liquid Clustering 54
Optimize 55
Predictive optimization 55
Applying our learning 55
Technical requirements 55
Project: Streaming Transactions 56
Project: Favorita Store Sales - Time Series Forecasting 64
Project: Retrieval Augmented Generation Chatbot 65
Project: Multilabel Image Classification 71
Summary 74
Questions 74
Answers 75
Further reading 75

Part 2: Heavily Project Focused

4

Getting to Know Your Data 79

Improving data integrity with DLT 80
Monitoring data quality with Databricks Lakehouse Monitoring 82
Mechanics of Lakehouse Monitoring 82
Visualization and alerting 82
Creating a monitor 83